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ODD END SEMESTER EXAMINATION, DECEMBER 2025

Name of the Course: B. Tech Biotechnology

Name of the Paper: Bioinformatics

Semester: V

Paper Code: TBT 504

Time: 3 hrs.

Maximum Marks: 100

Note:

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question
- Total marks in each main question are twenty.
- Each question carries 10 marks

Q1	20 Marks	COs
(a) Elaborate on the milestones in the development of bioinformatics. Discuss in detail about the Human Genome Project.		CO1, CO3, CO4
(b) Discuss the different types of biological databases used in Bioinformatics, giving examples for each. Explain the importance of primary, secondary, and composite databases with examples.		
(c) Define sequence alignment and discuss the differences between global and local alignment. Explain the practical implications of each approach in analyzing biological sequences.		
Q2	20 Marks	
(a) Define and differentiate between genomics, transcriptomics, proteomics, and metabolomics. Discuss their tools, technologies, and applications in functional biology and disease analysis.		CO2, CO4
(b) Compare different methods used for MSA, such as Clustal Omega, MUSCLE, and T-Coffee. Analyze their applications in phylogenetics and protein structure prediction.		
(c) Propose an omics-based experimental design to identify genes responsible for drought tolerance in plants. Outline the steps from sample collection to bioinformatics analysis and gene validation.		
Q3	20 Marks	
(a) Design a computational workflow to analyze multiple protein sequences and		

	construct a phylogenetic tree.	
(b)	Apply the Needleman–Wunsch algorithm to determine the optimal global alignment between the sequences S1 = ACTGAC and S2 = ACAGTC using the scoring scheme <i>Match</i> = +1, <i>Mismatch</i> = -1, and <i>Gap</i> = -2. Construct the scoring matrix.	CO3, CO4
(c)	Describe the FASTA algorithm used for sequence alignment. Explain how it differs from BLAST in terms of speed, sensitivity, and search strategy.	
Q4	20 Marks	
(a)	Explain the significance of scoring matrices in sequence alignments. Compare PAM and BLOSUM matrices, explaining their development, principles, and when to use each.	CO3, CO4
(b)	A researcher wants to identify potential drug targets in a new bacterial pathogen. Explain how NCBI, UniProt, and KEGG databases can be used to obtain genomic, proteomic, and pathway information for target identification.	
(c)	Discuss the major methods of phylogenetic tree construction, including UPGMA, Neighbor-Joining, Maximum Parsimony, and Maximum Likelihood. Explain briefly how bootstrapping validates tree reliability.	
Q5	20 Marks	
(a)	Evaluate the role of molecular visualization tools such as RasMol and PyMOL in understanding protein–ligand interactions.	CO5
(b)	Define pharmacophore modeling and discuss its importance in novel drug design. Explain the difference between structure-based and ligand-based drug design.	
(c)	Discuss the different computational approaches used to predict protein structures, including homology modeling, threading (fold recognition), and ab initio modeling.	